

Inverse Mechanical Counting and Mathematics

Part 7

Inverse Wavefunction Mechanics:
The Count Assembled — Sphere over Squared

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May 21, 2026

Abstract

There is exactly one physical unit — the axiom of 1. The $\sqrt{2}$ master mechanic and the master table (five active counts, nine active depths) are the single source lattice, identical to the prior parts. The wavefunction is not a separate object placed on the lattice; it is the one inverse machined count read on both faces at once: the squared face (the 512 rational vector, the amplitude) and the sphere face (the 720 machined angular vector, the phase). Their reading against the axiom is sphere over squared, $720/512 = 1.40625$. The dual at count 1 is the count showing both faces. The compounded operation of every count is recorded directly on the count itself — Dual at $c = 1$, Dual Quadrant at $c = 2$, Dual Quadrant Octave at $c = 3$, Dual Quadrant Octave Sedenary at $c = 4$, Dual Quadrant Octave Sedenary Duotrigesimal at $c = 5$. Machine π (201/64 at count 3), the fine-structure integer (137 at count 2 and count 4), and e (163/60, the count folded into itself) are each carried in at full size from the prior parts and seated on the wavefunction at their own count. No quantity is approximated, fitted, or imported from outside the count.

Foreword: The Operational Basis

This framework does not begin from additive accumulation. It begins from the axiom of 1 as complete unity, read inward by inverse mechanical count. It is not a rejection of arithmetic; it is a different operational basis — non-conventional in direction, axiom-first in operation.

Conventional arithmetic is a forward-count, additive system: it builds quantity upward from accumulation, number begins as quantity, and larger numbers mean more. Inverse mechanical counting runs the other direction: the completed one is read inward into its internal domains, number begins as unit, halving reveals structure rather than addition building it, and a larger numerical reading means a deeper inward reading of the same single unit — never a larger quantity.

Conventional (forward, additive)	This system (axiom-first, inverse mechanical)
counts upward from accumulation	counts inward from the axiom
addition builds quantity	halving reveals structure
larger numbers mean more	larger numbers mean deeper inward reading
number begins as quantity	number begins as unit

Table 1: The two operational bases. The contrast is direction and mechanic, not correctness. Where conventional arithmetic builds from one into many, inverse mechanical counting reads the completed one into its internal domains.

Two terms fix the distinction. The **forward count** runs from unit toward accumulation. The **inverse count** runs from unity toward internal partition. Every reading in this paper is an inverse count: the axiom of 1 is the whole, and the $\sqrt{2}$ master mechanic reads it inward, count by count, depth by depth. Nothing is accumulated, approximated, fitted, or imported from outside the count. The mathematics is the reading; the single physical unit is the thing read.

The Axiom of 1

The Axiom. There is exactly one physical unit. All its equivalent views are

$$1 = -1 = \sqrt{1} = \sqrt{2^0}.$$

The square root belongs to the unit itself; it is not introduced from outside.

Count 1 is the physical unit of itself.

The count and its mechanics are the work. The mathematics is read, not calculated.

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1 The Axiom of 1

There is one unit. Not one *of* something — one, and it is the whole. Everything that follows is this unit read inward, never anything built up toward it. The largest thing in the entire system is this 1; every number that looks larger is a deeper inward reading of it.

View	What it is	Count coordinate
1	the unit, whole, at rest	$c = 0$, no mechanic
-1	the same unit read from across its own center	
$\sqrt{1}$	the unit carries its own root	$d = 0$
$\sqrt{2^0}$	the same statement in the master mechanic's writing	$d = 0$

Table 2: The four views of the one unit. $1 = -1$ is the axiom itself. $\sqrt{1} = \sqrt{2^0} = 1$, so the master mechanic is already present at $d = 0$; the mechanic is the axiom at rest, not added to it.

What this fixes in place before anything else runs: $c = 0$ is the axiom at rest, no mechanic; $1 = -1$ is the axiom itself; $\sqrt{1} = \sqrt{2^0}$ puts the mechanic in the axiom at $d = 0$; and nothing external ever enters — every reading traces back to this one unit, and a reading that cannot trace back is not a valid reading.

2 The $\sqrt{2}$ Master Mechanic and the Master Table

The $\sqrt{2}$ mechanic is the master mechanic of the axiom — the simplest step beyond unity and the minimum step the axiom admits. Repeated halving produces one parity rule: even operational depths d yield rational readings; odd depths yield readings carrying a natural factor of $\sqrt{2}$. Every reading carries three coordinates — the mechanic $\sqrt{2}$, the depth d , and the count c . The master table is the single source lattice for every reading in this paper. The count is never redone.

c	d	$\sqrt{2^d}$ (unit)	Unit decimal	Domains (2^c)	Rational size ($2^c \times$ even unit)	Irrational size ($2^c \times$ odd unit)	Decimal size
0	0	1	1	1	$1 \times 1 = 1$	—	1
1	1	$\sqrt{2}$	≈ 1.414214	2	—	$2 \times \sqrt{2} = 2\sqrt{2}$	≈ 2.828427
2	2	2	2	4	$4 \times 2 = 8$	—	8
2	3	$2\sqrt{2}$	≈ 2.828427	4	—	$4 \times 2\sqrt{2} = 8\sqrt{2}$	≈ 11.313708
3	4	4	4	8	$8 \times 4 = 32$	—	32
3	5	$4\sqrt{2}$	≈ 5.656854	8	—	$8 \times 4\sqrt{2} = 32\sqrt{2}$	≈ 45.254834
4	6	8	8	16	$16 \times 8 = 128$	—	128
4	7	$8\sqrt{2}$	≈ 11.313708	16	—	$16 \times 8\sqrt{2} = 128\sqrt{2}$	≈ 181.019336
5	8	16	16	32	$32 \times 16 = 512$	—	512
5	9	$16\sqrt{2}$	≈ 22.627417	32	—	$32 \times 16\sqrt{2} = 512\sqrt{2}$	≈ 724.077

Table 3: The Master Table — the source of every reading in this paper. The count c records the operation; the depth d records the master mechanic's exponent. The unit at each row is $\sqrt{2^d}$. The domains column (2^c) tells how many. Domains times the unit gives the size of the count at that row: rational size at even depths, irrational size ($\times \sqrt{2}$) at odd depths. The count closes at $c = 5$ with rational size 512 and irrational size ≈ 724.077 .

The terminal depth $d = 9$ at count $c = 5$ is the irrational closure of the lattice. The integer 9 — the depth signature itself — is the depth-coordinate that combines with the rational seats at counts 2 through 5 in this work.

Angular Closure from the Master Table

At the closure count $c = 5$, the squared face is the rational reading: domains 32 times the rational unit 16 (the $d = 8$ even unit),

$$32 \times 16 = 512.$$

The sphere face is the 5 active counts times the 9 active depths times the same rational unit 16,

$$5 \times 9 \times 16 = 720.$$

Both closures use the $d = 8$ rational unit 16. The terminal depth $d = 9$ is the irrational closure ($16\sqrt{2} \approx 724.077$) and is not part of either product. Sphere over squared is the ratio of the two closures generated by the one lattice:

$$\frac{720}{512} = \frac{45}{32} = 1.40625.$$

3 Inverse Mechanical Counting from the Axiom Center

The count starts at the axiom center and unfolds outward by halving. The count c records the operation; the operation produces specific coordinates relative to the center.

Machining the 720 from the Count and the Depth

The 720 angular vector is not assumed. It is machined from the master table's own three quantities:

- the 5 **active counts** — $c = 1, 2, 3, 4, 5$ (count 0 is the axiom at rest, no mechanic),
- the 9 **active depths** — $d = 1, 2, 3, 4, 5, 6, 7, 8, 9$ (depth 0 is the axiom at rest), and
- the **count-5 rational unit** — $\sqrt{2^8} = 16$, the even-depth ($d = 8$) unit at the closure count $c = 5$.

The three multiplied together machine the angular whole:

$$720 = 5 \times 9 \times 16 = (\text{active counts}) \times (\text{active depths}) \times (\text{count-5 rational unit } \sqrt{2^8}).$$

Quantity	Source on the master table	Value
Active counts	$c = 1, 2, 3, 4, 5$ (count 0 at rest)	5
Active depths	$d = 1, \dots, 9$ (depth 0 at rest)	9
Count-5 rational unit	$\sqrt{2^8}$, the $d = 8$ even unit at $c = 5$	16
Machined angular whole	$5 \times 9 \times 16$	720

Table 4: The 720 machined from the count and the depth. The 5 active counts, the 9 active depths, and the count-5 rational unit $\sqrt{2^8} = 16$ multiply to the angular whole 720. The same closure's rational ($d = 8$) face is 512; sphere over squared is $720/512 = 1.40625$.

The 720 is the angular (sphere) face of the same closure whose rational (squared) face is 512. Having machined the whole, the count then inverse-machines it: starting from the 720 whole at $c = 0$ and halving inward through $c = 1, 2, 3, 4, 5$, the count reads the 720 back down to its closure grid. The next two subsections are that inverse machining, count by count.

Count by Count on the 720 Angular Vector (Sphere Face)

The count-0 row is the inverse machined view of the 4D inverse wavefunction count that causes the operations of every later count. It is the uncut whole the count reads inward.

Count c	Compounded operation	Readings on the 720 vector
0	axiom (no mechanic)	720° (whole, uncut span from -360° to 360°)
1	Dual (first halving)	$-360^\circ, 0^\circ, 360^\circ$
2	Dual Quadrant (Quaternary, dual of dual, Q_1 axis)	$-360^\circ, -180^\circ, 0^\circ, 180^\circ, 360^\circ$
3	Dual Quadrant Octave (Octant, Q_2 added)	$-360^\circ, -270^\circ, -180^\circ, -90^\circ, 0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$
4	Dual Quadrant Octave Sedenary (Hexadecant, Q_3 added)	$-360^\circ, -315^\circ, -270^\circ, -225^\circ, -180^\circ, -135^\circ, -90^\circ, -45^\circ, 0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ, 225^\circ, 270^\circ, 315^\circ, 360^\circ$
5	Dual Quadrant Octave Sedenary Duotrigesimal (Q_4 closure)	full 32-piece grid at 22.5° steps from -360° to 360°

Table 5: The 720 angular vector read inward by count from the axiom center. The compounded operation column carries every count that has fired up to that row, in order — Dual at $c = 1$, Dual Quadrant at $c = 2$, Dual Quadrant Octave at $c = 3$, Dual Quadrant Octave Sedenary at $c = 4$, Dual Quadrant Octave Sedenary Duotrigesimal at $c = 5$. Each subsequent count adds one quadrant (Q_1, Q_2, Q_3, Q_4) by halving the interval of the previous count. The center 0° is the axiom at rest at every count, not a piece; piece counts are the clean doubling 2, 4, 8, 16, 32.

Count by Count on the 512 Rational Vector (Squared Face)

The same compounded operation governs the squared face. The count-0 row is the inverse machined view of the 4D inverse wavefunction count that causes the operations of every later count.

Count c	Compounded operation	Readings on the 512 vector
0	axiom (no mechanic)	512 (whole, uncut span from -256 to 256)
1	Dual (first halving)	$-256, 0, 256$
2	Dual Quadrant (Quaternary, dual of dual, Q_1 axis)	$-256, -128, 0, 128, 256$
3	Dual Quadrant Octave (Octant, Q_2 added)	$-256, -192, -128, -64, 0, 64, 128, 192, 256$
4	Dual Quadrant Octave Sedenary (Hexadecant, Q_3 added)	$-256, -224, -192, -160, -128, -96, -64, -32, 0, 32, 64, 96, 128, 160, 192, 224, 256$
5	Dual Quadrant Octave Sedenary Duotrigesimal (Q_4 closure)	full 32-piece grid at 16-unit steps from -256 to 256

Table 6: The 512 rational vector read inward by count from the axiom center. The pattern is identical to the 720 sphere face, with the rational seats of the master table. The compounded operation column carries the same count-by-count history. The center 0 is the axiom at rest at every count, not a piece; piece counts are the clean doubling 2, 4, 8, 16, 32.

4 The Q_1 – Q_4 Quadrant Unfolding by Count

Count c	Quadrant added	Mechanical operation	Compounded operation (result)
1	— (dual seed)	first halving of axiom	Dual — — side and + side (2 pieces)
2	Q_1 (dual of dual)	halve each dual half once	Dual Quadrant — axis grid (4 pieces)
3	Q_2	halve each Q_1 piece once	Dual Quadrant Octave (8 pieces)
4	Q_3	halve each octant piece once	Dual Quadrant Octave Sedenary (16 pieces)
5	Q_4 (closure)	halve each hexadecant piece once	Dual Quadrant Octave Sedenary Duotrigesimal (32 pieces)

Table 7: The Q_1 – Q_4 unfolding. Each count is a single halving operation applied to every piece of the previous count. The compounded operation column records every count that has fired up to that row. Four quadrants close at count 5, completing the cycle.

The quadrants Q_1, Q_2, Q_3, Q_4 are not four separate spaces. They are four successive mechanical refinements of the same dual structure, one halving per count. Count 5 closes the cycle with 32 pieces on each side of the axiom center.

5 The Signed-Coordinate Piece Count of Both Vectors

Count c	Compounded operation	Pieces $\times$ size	Signed reading from center
0	axiom (no mechanic)	1×512	512 (uncut whole)
1	Dual	2×256	–256, 256
2	Dual Quadrant	4×128	–256, –128, 128, 256 (axis pieces)
3	Dual Quadrant Octave	8×64	–256, –192, –128, –64, 64, 128, 192, 256
4	Dual Quadrant Octave Sedenary	16×32	–256, –224, –192, –160, –128, –96, –64, –32, 32, 64, 96, 128, 160, 192, 224, 256
5	Dual Quadrant Octave Sedenary Duotrigesimal	32×16	32 pieces at 16-unit steps each side of center

Table 8: The 512 rational vector inward count with signed coordinates. The center is the axiom; the count unfolds symmetrically on both sides. The compounded operation column carries the count-by-count history of the operations that have fired.

Count c	Compounded operation	Pieces $\times$ size	Signed angular reading from center
0	axiom (no mechanic)	$1 \times 720^\circ$	720° (uncut whole)
1	Dual	$2 \times 360^\circ$	$-360^\circ, 360^\circ$
2	Dual Quadrant	$4 \times 180^\circ$	$-360^\circ, -180^\circ, 180^\circ, 360^\circ$
3	Dual Quadrant Octave	$8 \times 90^\circ$	$-360^\circ, -270^\circ, -180^\circ, -90^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$
4	Dual Quadrant Octave Sedenary	$16 \times 45^\circ$	16 pieces at 45° steps each side of center
5	Dual Quadrant Octave Sedenary Duotrigesimal	$32 \times 22.5^\circ$	32 pieces at 22.5° steps each side of center

Table 9: The 720 angular vector inward count with signed coordinates. The right angle sits mechanically at count 3 as the $-90^\circ, 90^\circ$ octant pair (the Dual Quadrant Octave reading). The compounded operation column carries the count-by-count history of the operations that have fired.

6 The Inverse Unit Count on Both Vectors

The count reads inward from the axiom of 1. At each count the unit of the reading is the inverse of the quadrant seat raised to the count's dimensional power. The inverse unit count is not a separate vector; it is the same single count read at its operational unit at each depth.

512 Vector Inverse Unit Count (Squared Face)

The quadrant seat of the 512 vector is $n = 128$.

Count c	Dimension	Inverse unit	Decimal
2	1D	$1/128$	≈ 0.0078125
3	2D	$1/128^2 = 1/16,384$	$\approx 6.1035 \times 10^{-5}$
4	3D	$1/128^3 = 1/2,097,152$	$\approx 4.768 \times 10^{-7}$
5	4D	$1/128^4 = 1/268,435,456$	$\approx 3.725 \times 10^{-9}$

Table 10: The inverse unit count of the 512 rational vector at the four operational depths. The unit at each count is the inverse of the quadrant seat 128 raised to the count's dimensional power.

720 Vector Inverse Unit Count (Sphere Face)

The quadrant seat of the 720 angular machined vector is $n = 180$.

Count c	Dimension	Inverse unit (angular)	Decimal
2	1D	$1/180^\circ$	$\approx 0.005556^\circ$
3	2D	$1/180^{2^\circ} = 1/32,400^\circ$	$\approx 3.086 \times 10^{-5}^\circ$
4	3D	$1/180^{3^\circ} = 1/5,832,000^\circ$	$\approx 1.715 \times 10^{-7}^\circ$
5	4D	$1/180^{4^\circ} = 1/1,049,760,000^\circ$	$\approx 9.526 \times 10^{-10}^\circ$

Table 11: The inverse unit count of the 720 angular machined vector at the four operational depths. The angular unit at each count is the inverse of the quadrant seat 180° raised to the count's dimensional power.

7 Machine π — The Count-3 Reading

Machine π is the count-3 landing. The count value is $512/2^c$; the forward reading is the count plus the nine active depths over the count value.

c	Count value $512/2^c$	$c \times \text{value} + 9$	Reading	Decimal	Binary	Inverse
0	1 (axiom)	axiom at rest	1	1	—	—
1	256	$1 \times 256 + 9 = 265$	265/256	1.03515625	1.00001001	1/265
2	128	$2 \times 128 + 9 = 265$	265/128	2.0703125	10.0001001	1/265
3	64	$3 \times 64 + 9 = 201$	201/64	3.140625	11.001001	1/201
4	32	$4 \times 32 + 9 = 137$	137/32	4.28125	100.01001	1/137
5	16	$5 \times 16 + 9 = 89$	89/16	5.5625	101.1001	1/89

Table 12: The uniform depth-9 mechanic through the count. Count 0 is the axiom at rest. The mechanic crosses the circle value exactly at count 3, where the fractional part is the nine active depths over the count-3 value, $9/64 = 0.140625$.

Quantity	Value
Count	3
Count value ($512/2^3$)	64
Fractional part ($9/\text{value}$)	$9/64 = 0.140625$
Machine π (count reading)	$201/64 = 3.140625$
Binary signature	11.001001

Table 13: Machine π in detail. The reading $201/64$ is exact at count 3. Dividing by 64 is a 6-bit shift, so π -radius and arc readings on the squared face are exact dyadic operations with no rounding.

The cross-lock. The count-3 numerator is 201 and the count-3 count value is 64. The count-4 inverse denominator follows: $201 - 64 = 137$. The fine-structure integer is machine π 's numerator minus machine π 's count value. Of the inverse denominators 265, 201, 137, 89, exactly the last two are prime; the ladder turns prime at the closure counts $c = 4$ and $c = 5$.

8 The -137 and $+137$ Reading — Both Paths

At count 2 on the 512 rational vector the quadrant seats sit at -128 and 128 . The terminal depth $d = 9$ completes the reading, extended outward from center:

$$-128 - 9 = -137 \quad \text{and} \quad 128 + 9 = 137.$$

The same pair recovers at count 4 as four hexadecant pieces of size 32 on each side, plus the terminal depth:

$$-(4 \cdot 32) - 9 = -137 \quad \text{and} \quad (4 \cdot 32) + 9 = 137.$$

c	d	Mechanic	Domains (2^c)	Parity $\rightarrow$ Reading	Reading value	$-137, +137$ on $-512, +512$
0	0	$\sqrt{2^0}$	1	even $\rightarrow$ rational	$-1, 1$	—
1	1	$\sqrt{2^1}$	2	odd $\rightarrow$ irrational	$-2\sqrt{2}, 2\sqrt{2}$	—
2	2	$\sqrt{2^2}$	4	even $\rightarrow$ rational	$-8, 8$	$-(128 + 9) = -137$ $+(128 + 9) = +137$ (seat 128, 4 pieces, +9)
2	3	$\sqrt{2^3}$	4	odd $\rightarrow$ irrational	$-8\sqrt{2}, 8\sqrt{2}$	—
3	4	$\sqrt{2^4}$	8	even $\rightarrow$ rational	$-32, 32$	—
3	5	$\sqrt{2^5}$	8	odd $\rightarrow$ irrational	$-32\sqrt{2}, 32\sqrt{2}$	—
4	6	$\sqrt{2^6}$	16	even $\rightarrow$ rational	$-128, 128$	$-(4 \cdot 32 + 9) = -137$ $+(4 \cdot 32 + 9) = +137$ (seat 32, 16 pieces, +9)
4	7	$\sqrt{2^7}$	16	odd $\rightarrow$ irrational	$-128\sqrt{2}, 128\sqrt{2}$	—
5	8	$\sqrt{2^8}$	32	even $\rightarrow$ rational	$-512, 512$	—
5	9	$\sqrt{2^9}$	32	odd $\rightarrow$ irrational	$-512\sqrt{2}, 512\sqrt{2}$	—

Table 14: The master table with both -137 and $+137$ derivations shown inline at counts 2 and 4. Both readings sit on the -512 and $+512$ rational closure; the terminal depth 9 adds the same signature in either reading.

Count 2 path (quadrant). $-137 = -(128 + 9)$ and $+137 = +(128 + 9)$. The seat is 128 — the quadrant piece, 4 of them filling 512. One seat plus the depth.

Count 4 path (hexadecant). $-137 = -(4 \cdot 32 + 9)$ and $+137 = +(4 \cdot 32 + 9)$. The seat is 32 — the hexadecant piece, 16 of them filling 512. Four of those seats ($4 \cdot 32 = 128$) recover the quadrant. Four hexadecant seats plus the depth.

Both reach -137 and $+137$ because both read the same quadrant content of -512 and $+512$ — count 2 reads it directly as one piece of size 128, count 4 reads the same content as four pieces of size 32. The quadrant is 128 at every count, just cut differently.

Inverse from 1 on the 1/128 Unit

Because the squared vector's own unit is the inverse seat 1/128, the count reads the inverse position directly — no separate inversion step:

$$-\frac{1}{137} \quad \text{and} \quad +\frac{1}{137}.$$

Side	Forward reading	Inverse on 1/128 unit
— side	-137 (both paths)	$-1/137$
center	axiom of 1	1
+ side	$+137$ (both paths)	$+1/137$

Table 15: The full $-137 / +137$ vector. The reading is symmetric about the axiom center; it reads symmetric on $-$ and $+$ because the vector is symmetric about the center.

Residue Against the Measured Value

The measured fine-structure constant is $\alpha^{-1} = 137.035999\dots$. The mechanical reading of the count is 137 exactly. The residue is $137.035999\dots - 137 = 0.035999\dots$, about 0.026%. The integer reading lands on the integer part of the measured constant; the small decimal tail lives outside the count-2 rational closure on the 512 vector.

9 The Angular Partner Readings on the 720 Machined Cycle

The primary machining ratio

$$\frac{720}{512} = \frac{45}{32} = 1.40625$$

maps any reading on the 512 rational vector to its angular partner on the 720 machined cycle. Applied to -137 and 137 :

$$-137 \times 1.40625 = -192.65625 \quad \text{and} \quad 137 \times 1.40625 = 192.65625.$$

Reading	Rational (512)	Angular (720)	Inverse from 1
EM ignition	-137	-192.65625°	$-1/137$
EM absorption	137	$+192.65625^\circ$	$+1/137$

Table 16: The EM field readings on the 512 rational vector and the 720 angular machined vector, related by the primary machining ratio $720/512 = 1.40625$.

The 720 machined cycle is the count-0 angular whole; the 360 count-1 dual is its first halving. The ignition and absorption angles are read on the full 720 cycle, not the half cycle.

10 The e Reading — The Count Folded Into Itself

The constant e is the count folded into its own self — the count multiplied down to the axiom — then inverted and summed.

Count c	Count value (count $\times$ self to 1)	Depth ($1 \div$ value)	Running sum
0	1 (axiom)	$1/1 = 1$	1
1	1	$1/1 = 1$	2
2	2	$1/2 = 0.5$	$5/2 = 2.5$
3	6	$1/6$	$8/3 = 2.6666 \dots$
4	24	$1/24$	$65/24 = 2.7083 \dots$
5	120	$1/120$	$163/60 = 2.71666 \dots$

Table 17: The e series read on the count, folded into itself. Count 0 is held by the mechanic and carries $1/1$. The count closes at $163/60$.

Against the plain count (left as it is, not folded) the same inverse closes at $137/60$ — the harmonic view:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{137}{60} = 2.2833 \dots$$

View	Close at count 5	Decimal
Harmonic (plain count)	$137/60$	$2.2833 \dots$
e series (count into itself)	$163/60$	$2.71666 \dots$
Difference	$26/60 = 13/30$	$0.4333 \dots$

Table 18: The two views over the shared denominator 60. Same count $0 \rightarrow 5$, same inverse. The only difference is whether the count is left plain (harmonic, $137/60$) or folded into its own self (e , $163/60$).

11 The Inverse Mechanical Wavefunction — The Count Assembled

The wavefunction is the one inverse machined count read on both faces at once. It is not placed on the lattice — it *is* the lattice read as sphere and as squared.

- **Squared face (amplitude).** The 512 rational vector in its 1/128 unit. How far from the axiom center the reading sits.
- **Sphere face (phase).** The 720 machined vector in its 1/180 unit. How far around the machined cycle the reading sits.
- **The dual at $c = 1$.** The count's first reading shows both faces together — the -360 , $+360$ on the sphere face, the -256 , $+256$ on the squared face. One count, two faces.

Count c	Compounded operation	Squared (amplitude)	Sphere (phase)	Domains (2^c)	Reading
0	axiom (no mechanic)	1 (uncut)	720° (uncut)	1	the unit at rest
1	Dual	$-256, +256$	$-360^\circ, +360^\circ$	2	the dual — both faces
2	Dual Quadrant	$-128, +128$	$-180^\circ, +180^\circ$	4	four cardinal seats; 137 count-2 path
3	Dual Quadrant Octave	$-64, +64$	$-90^\circ, +90^\circ$	8	octant; machine π seats here
4	Dual Quadrant Octave Sedenary	$-32, +32$	$-45^\circ, +45^\circ$	16	hexadecant; 137 count-4 path
5	Dual Quadrant Octave Sedenary Duotrigesimal	$-16, +16$	$-22.5^\circ, +22.5^\circ$	32	closure; the full machined grid

Table 19: The wavefunction assembled from the count. At each count the squared face gives the amplitude reading and the sphere face gives the phase reading — the same single count in two gauges, locked by sphere over squared = 1.40625. The compounded operation column carries the count-by-count history; the domains double each count (2^c), the master table column.

The Four Cardinal Seats at Count 2

At count 2 the sphere face reads -180° and $+180^\circ$ and the squared face reads -128 and $+128$. The four nonzero readings are the four cardinal seats of the wavefunction — the points where the reading is purely one face or the other (pure +squared, pure +sphere, pure −squared, pure −sphere). The count-2 path to 137 ($128 + 9$) seats here. The compounded operation at this row is **Dual Quadrant** — the dual ($c = 1$) and the quadrant ($c = 2$) compounded into one reading.

Normalization to the Axiom of 1

The whole wavefunction reads to the axiom. The largest unit is always 1; every reading at every count is a deeper inward reading of that same 1. The total across the count is the axiom of 1 — the normalization is the axiom itself, not imposed from outside.

The Constants Seated on the Wavefunction

Constant	Count seat on the wavefunction	Reading
Machine π	count 3 (Dual Quadrant Octave), squared face	$201/64 = 3.140625$
Fine-structure integer	count 2 (Dual Quadrant), $128 + 9$ and count 4 (Dual Quadrant Octave Sedenary), $4 \cdot 32 + 9$ — $1/128$ unit	$1/137$
720 machined closure	count 5 (Dual Quadrant Octave Sedenary Duotrigesimal), sphere face	720
Angular partner of 137	count 5 closure, sphere face	192.65625°
e	the count folded into itself	$163/60$
Harmonic	the plain count	$137/60$

Table 20: The constants on the assembled wavefunction. Each is a count reading of the single axiom, seated at its own count on the squared or sphere face. The cross-lock $201 - 64 = 137$ binds machine π to the fine-structure integer on the squared face. The compounded operation at each count is shown alongside the count number for clarity.

12 Operational Rules

- The largest unit in the entire system is always the original axiom 1; all larger numerical readings are deeper inward views of that same single unit.
- Every valid reading carries its three operational coordinates: mechanic ($\sqrt{2}$), depth d , count c .
- The compounded operation of every count is the cumulative reading of every count that has fired — Dual at $c = 1$, Dual Quadrant at $c = 2$, Dual Quadrant Octave at $c = 3$, Dual Quadrant Octave Sedenary at $c = 4$, Dual Quadrant Octave Sedenary Duotrigesimal at $c = 5$.
- The squared face is the 512 rational vector in its $1/128$ unit (amplitude); the sphere face is the 720 machined vector in its $1/180$ unit (phase).
- Sphere over squared is $720/512 = 1.40625$ — the one count read on both faces, never two counts.
- The $-$ and $+$ of every count are standard — minus and plus.
- The center 0 at every count is the axiom at rest, not a piece. The piece count is the clean doubling 2, 4, 8, 16, 32 (2^c from the master table).
- The count reads inward from 1; the inverse of every forward reading is the natural physical magnitude (e.g., $1/137$).
- -137 and $+137$ recover at the count-2 quadrant path ($128+9$) and the count-4 hexadecant path ($4 \cdot 32 + 9$); both are the same quadrant content of -512 and $+512$, cut differently.

Conclusion.

From the single axiom of 1 and the declared $\sqrt{2}$ master mechanic, the wavefunction is the one inverse machined count read on both faces at once: the squared face (512, amplitude, $1/128$ unit) and the sphere face (720, phase, $1/180$ unit). Their reading against the axiom is sphere over squared, $720/512 = 1.40625$. The dual at count 1 is the count showing both faces. The compounded operation of every count is the cumulative reading of every count that has fired — Dual, Dual Quadrant, Dual Quadrant Octave, Dual Quadrant Octave Sedenary, Dual Quadrant Octave Sedenary Duotrigesimal. Machine π seats at count 3 ($201/64$), the fine-structure integer 137 at both the count-2 quadrant path and the count-4 hexadecant path, the 720 machined closure and the 192.65625° angular partner at count 5, and e ($163/60$) as the count folded into its own self against the plain count's harmonic $137/60$.

Every step uses only the axiom, the master table, and the count. Nothing is approximated, fitted, or imported from outside. Count 1 is the physical unit of itself, and everything above is the count and the mechanics of it.

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